IS51 · Machine Learning

Unit 3: Regression Analysis & Regularization

Handwritten Student Notebook

Syllabus Topics Covered:

CIE-1 AUTHORITATIVE SYLLABUS TOPICS COVERED Introduction to Regression: Dependent vs independent variables, Regression vs Classification, Performance evaluation metrics (MSE, RMSE, MAE, R^2, Adjusted R^2). Simple Linear Regression: Model assumptions, Ordinary Least Squares (O...

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Introduction to Regression Analysis

§ 1

Regression analysis models the functional relationship between one or more ***independent explanatory variables*** (features $\mathbf{X} = [x_1, \ldots, x_d]$) and a ***continuous numerical dependent variable*** (target Y).

Regression vs. Classification

- ***Target Domain:*** Classification outputs discrete categorical classes ($Y \in \{0, 1\}$ or $\{C_1, \ldots, C_k\}$); Regression outputs real-valued continuous numbers ($Y \in \mathbb{R}$).
- ***Evaluation Metrics:*** Classification uses Accuracy, Precision, Recall, F_1, and ROC-AUC; Regression uses Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2).

Quick Recall

Regression predicts continuous numerical quantities ($Y \in \mathbb{R}$). Evaluated via Mean Squared Error (MSE) and R^2.

Linear Regression: Ordinary Least Squares (OLS)

§ 2

Linear regression assumes a linear relationship between features and target.

Simple Linear Regression (Single Feature)

$$y = mx + c + \epsilon$$

Ordinary Least Squares (OLS) minimizes the Residual Sum of Squares (RSS):

$$RSS = \sum_{i=1}^{n} (y_i - (mx_i + c))^2$$

Taking partial derivatives with respect to m and c and setting them to zero yields the closed-form solutions:

OLS Closed-Form Formulas:

$$m = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}$$

$$c = \bar{y} - m\bar{x}$$

Multiple Linear Regression (Matrix Normal Equation)

For d features, $\hat{y} = \mathbf{X}\mathbf{w}$. The closed-form analytical solution that minimizes Mean Squared Error is the **Normal Equation**:

$$\mathbf{w} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})(y_i - \bar{y})$	$(y_i - \bar{y})^2$
1	2	-2	-2	4	$+4$	4
2	3	-1	-1	1	$+1$	1
3	5	0	$+1$	0	0	1
4	4	$+1$	0	1	0	0
5	6	$+2$	$+2$	4	$+4$	4
Total		**0**	**0**	**10**	**9**	**10**

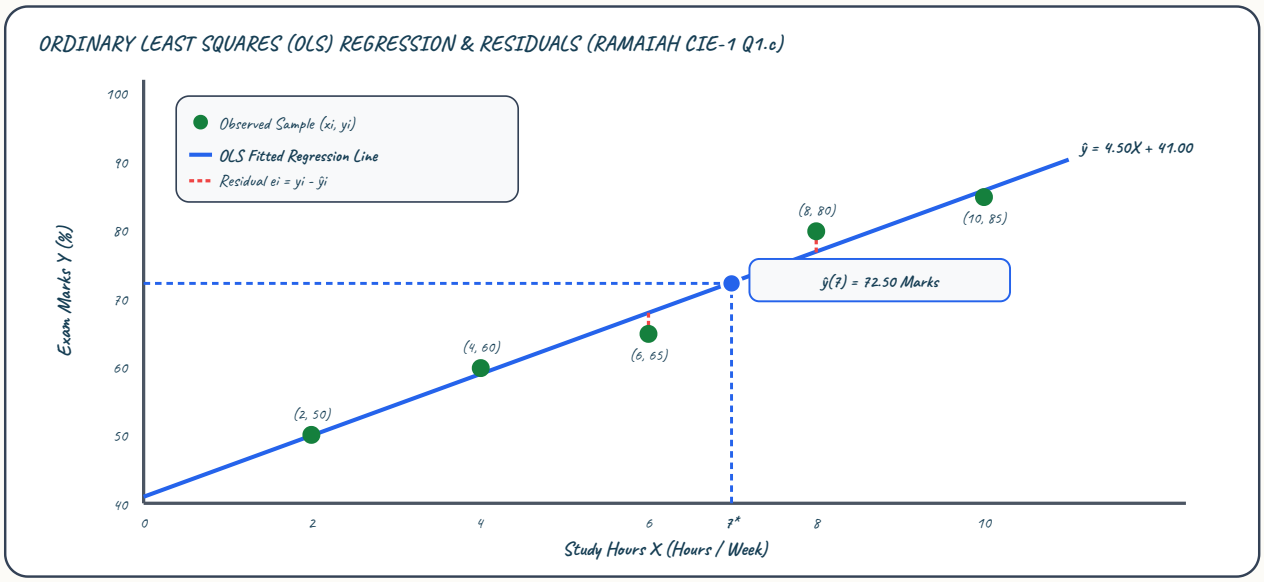


Figure 3.1: Ordinary Least Squares (OLS) Regression Fit & Residuals (Ramaiah CIE-1 Q1.c Solved)

Study Hours (X) vs Exam Marks (Y) yields optimal least-squares line $\hat{Y} = 4.50X + 41.00$. Vertical red lines depict residuals $\mathbf{e}_i = y_i - \hat{y}_i$. Target prediction for $X = 7$ hours is $\hat{Y}(7) = 72.50$ marks.

> **[Interactive Visualizer Online]:** *Interactive Linear Regression & Error Minimizer. Explore the step-by-step visualizer on the live course notes page.*

Quick Recall

OLS slope $m = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}$ and $c = \bar{y} - m\bar{x}$. Matrix Normal Equation: $\mathbf{w} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$. $R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$.

Multiple Linear Regression & Normal Equations

§ 3

When a continuous outcome y depends on multiple predictors $x_1, x_2, \ldots, x_p$, **Multiple Linear Regression** generalizes simple OLS into matrix algebra.

Matrix Formulation of Multiple Regression:

For n observations and p features, the model is:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $\mathbf{y} \in \mathbb{R}^{n\times 1}$, design matrix $\mathbf{X} \in \mathbb{R}^{n\times(p+1)}$ (with leading column of 1s for intercept), coefficient vector $\boldsymbol{\beta} \in \mathbb{R}^{(p+1)\times 1}$, and residuals $\boldsymbol{\epsilon} \in \mathbb{R}^{n\times 1}$.

The Normal Equations (Closed-Form Analytical Solution)

Minimizing the sum of squared errors $S(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ by setting the gradient $\mathrm{del}_{\beta} S = \mathbf{0}$ yields the classic **Normal Equations**:

$$\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y} \implies \boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Assumption & Invertibility:

For $(\mathbf{X}^T\mathbf{X})^{-1}$ to exist, matrix $\mathbf{X}$ must have full column rank $(p+1)$. If features are linearly dependent (exact multicollinearity), $(\mathbf{X}^T\mathbf{X})$ becomes singular and cannot be inverted without regularization (Ridge regression).

Worked Numerical: Two-Predictor House Price Prediction (SEE 2026 Qs.a)

Predict house price (Y, in Lakhs) from Size (X_1, hundreds of sq.ft) and Bedrooms (X_2):

House	Size (X_1)	Bedrooms (X_2)	Price (Y, Lakhs)
H1	12	2	50
H2	15	3	65
H3	18	3	80
H4	20	4	90

Solving via least squares yields estimated model: $\hat{Y} = -6.67 + 4.17X_1 + 3.33X_2$.

Holding bedrooms fixed, each additional 100 sq.ft of area adds Rs. 4.17 Lakhs to the predicted price.

Polynomial Regression & Bias-Variance Tradeoff

§ 4

When the relationship between features and target is non-linear (e.g., parabolic or cyclical), fitting a straight line causes severe **underfitting (high bias)**. **Polynomial Regression** extends linear regression by introducing powers of features as new independent variables:

$$\hat{y} = w_0 + w_1 x + w_2 x^2 + w_3 x^3 + \cdots + w_d x^d$$

Crucially, polynomial regression remains a **linear model** because it is linear in its parameters $\mathbf{w}$, and can be solved directly using the standard OLS Normal Equation over the transformed feature matrix $\mathbf{\Phi}(X)$.

The Bias-Variance Dilemma Across Polynomial Degrees

Degree (d)	Model Behavior	Bias vs. Variance	Remedy
Degree 1 (Linear)	Underfitting: straight line fails to capture curvature	High Bias, Low Variance	Increase polynomial degree
Degree 2–3 (Optimal)	Optimal generalization: captures underlying trend without fitting noise	Balanced Bias and Variance	Cross-validation model selection
Degree 15+ (Excessive)	Overfitting: oscillates wildly to pass through every individual noise point	Low Bias, High Variance	Regularization (L_2 Ridge / L_1 Lasso)

Quick Recall

Polynomial regression models non-linear phenomena via $\sum w_k x^k$. It is linear in parameters. High degree polynomials cause overfitting (high variance), mitigated via Ridge/Lasso regularization.

Regularization in Regression: Ridge (L2) & Lasso (L1)

§ 5

When features are collinear or numerous ($pppproxn$), Ordinary Least Squares suffers from high variance (overfitting). **Regularization** constrains parameter magnitudes by adding a penalty term to the loss function.

Property	Ordinary Least Squares (OLS)	Ridge Regression (L_2)	Lasso Regression (L_1)
Penalty Term	None	$\lambda \sum_{j=1}^{p} \beta_j^2 = \lambda \|\boldsymbol{\beta}\|_2^2$	$\lambda \sum_{j=1}^{p} \|\beta_j\| = \lambda \|\boldsymbol{\beta}\|_1$
Objective Function	$\min \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$	$\min \left[\text{MSE} + \lambda \|\boldsymbol{\beta}\|_2^2\right]$	$\min \left[\text{MSE} + \lambda \|\boldsymbol{\beta}\|_1\right]$
Analytical Solution	$\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$	$\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$	No closed form; solved via Coordinate Descent
Sparsity / Feature Selection	No	No (shrinks coefficients toward zero, never exactly 0)	**Yes** (sets coefficients to exactly zero, eliminating features)
Computational Benefit	Can be singular if multicollinear	Guarantees $(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})$ is invertible	Produces compact, interpretable models

Faculty Problem Note (Shruti Mam's Problem Set):

In Ridge regression, increasing λ increases bias but dramatically reduces model variance. For $\lambda = 0$, Ridge reverts to standard OLS. As $\lambda \to \infty$, all $\beta_j \to 0$, predicting the horizontal mean $\bar{Y}$.

Gradient Descent Optimization for Regression

§ 6

When the number of features p is large (e.g. $p > 10,000$), computing the analytical matrix inverse $(\mathbf{X}^T\mathbf{X})^{-1}$ with time complexity $O(p^3)$ is computationally prohibitive. **Gradient Descent** provides an iterative optimization alternative.

Batch, Stochastic, and Mini-Batch Gradient Descent

- **Batch Gradient Descent:** Computes the gradient of MSE over the entire dataset n at each step. Convergence is smooth and monotonic, but slow for large datasets.
- **Stochastic Gradient Descent (SGD):** Computes the gradient using a single random training sample at each step. Fast updates with noisy trajectory; can escape shallow local minima.

- **Mini-Batch Gradient Descent:** Computes the gradient over a small subset (e.g. $B = 32, 64, 128$). Combines the stability of batch descent with hardware vectorization speedups (GPU/SIMD).

Update Rule: $\beta^{(t+1)} = \beta^{(t)} - \alpha \text{del}_{\beta} J(\beta)$, where α is the learning rate hyperparameter.

Logistic Regression

§ 7

Despite its name, **Logistic Regression** is a **classification algorithm** used to estimate the probability that an instance belongs to a binary class ($Y \in \{0, 1\}$).

The Sigmoid (Logistic) Function:

Linear regression maps input features to unbounded real values: $z = \mathbf{w}^T \mathbf{x} \in (-\infty, +\infty)$. Logistic regression passes z through the **Sigmoid Activation Function** $\sigma(z)$ to squash outputs into valid probabilities in $(0, 1)$:

$$P(Y = 1 \mid \mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x})}}$$

Properties: $\sigma(0) = 0.5$, $\sigma(z) \to 1$ as $z \to +\infty$, and $\sigma(z) \to 0$ as $z \to -\infty$.

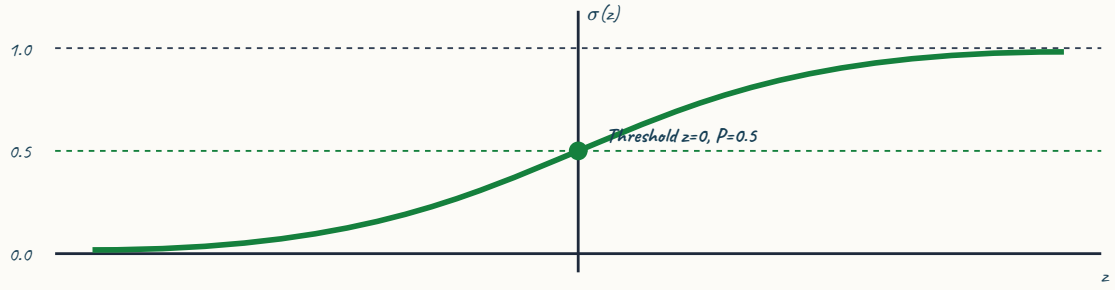


Figure 3.2: The logistic sigmoid curve mapping the real line into calibrated probabilities.

Odds Ratio and the Logit Function

The **Odds** of an event is the ratio of probability of occurrence to non-occurrence:

$$\text{Odds} = \frac{p}{1-p} = e^z$$

Taking the natural logarithm yields the **Logit**:

$$\ln\left(\frac{p}{1-p}\right) = z = w_0 + w_1 x_1 + \cdots + w_d x_d$$

Thus, logistic regression is linear in the log-odds space!

Loss Function: Binary Cross-Entropy (Log Loss)

We cannot use Mean Squared Error for logistic regression because squaring $\sigma(z)$ creates a non-convex loss surface plagued by local minima. Instead, we use Maximum Likelihood Estimation, leading to the convex **Binary Cross-Entropy Loss**:

$$J(\mathbf{w}) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \ln(\hat{p}_i) + (1-y_i)\ln(1-\hat{p}_i)\right]$$

Solved Problem 4 · Exam Level Logistic Classification

A trained binary logistic regression model for predicting cardiovascular risk has the following logit equation:

$$z = -3.0 + 0.5x_1 + 0.2x_2$$

Where x_1 is daily cigarette consumption and x_2 is patient body mass index (BMI).

1. Compute the logit value z for a patient with $x_1 = 4$ cigarettes and $x_2 = 10$ BMI units above baseline.
2. Compute the odds of developing cardiovascular disease for this patient.
3. Compute the predicted probability $P(Y = 1 \mid \mathbf{x})$.
4. Using a classification threshold of 0.5, predict the diagnostic class.

Step 1: Compute Logit z

$$z = -3.0 + 0.5(4) + 0.2(10) = -3.0 + 2.0 + 2.0 = +\mathbf{1.0000}$$

Step 2: Compute Odds

$$\text{Odds} = e^z = e^{1.0} \approx \mathbf{2.7183}$$

The patient is approximately 2.72 times more likely to be at risk than healthy.

Step 3: Compute Predicted Probability $\sigma(z)$

$$P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-1.0}} = \frac{1}{1 + 0.367879} = \frac{1}{1.367879}$$

Step 4: Classification Decision

Since $P(Y = 1 \mid \mathbf{x}) = 0.7311 \geq 0.50$, the patient is classified into ***Class 1 (High Risk)***.

Final Verified Results Logit z = +1.0000 | Odds = 2.7183 | P(Y=1) = 0.7311 (73.11%) → Classified as Positive (1)

Quick Recall

Logistic regression models probability via $\sigma(z) = \frac{1}{1+e^{-z}}$. Linear in log-odds $\ln \frac{p}{1-p} = \mathbf{w}^T\mathbf{x}$. Optimized via Binary Cross-Entropy (Log Loss).

Unit 3 Quick Revision Sheet (Regression Only)

High-yield formula cheat sheet and algorithm summaries for rapid CIE-1 exam review.

Simple OLS Regression

$$b_1 = \frac{n\sum XY - (\sum X)(\sum Y)}{n\sum X^2 - (\sum X)^2}$$

$$b_0 = \bar{Y} - b_1\bar{X}$$

$$\hat{Y} = b_0 + b_1 X$$

$$R^2 = 1 - \frac{SSE}{SST}$$

Multiple Linear Regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

$$\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

$$R^2_{\text{adj}} = 1 - \left[\frac{(1-R^2)(n-1)}{n-p-1}\right]$$

Ridge Regularization (L2)

Loss: $\text{MSE} + \lambda \sum \beta_j^2$

$$\boldsymbol{\beta}_{\text{Ridge}} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

Shrinks weights; never zero.

Lasso Regularization (L1)

Loss: $\text{MSE} + \lambda \sum |\beta_j|$

Enforces sparsity: sets weights to 0.

Performs feature selection.

Gradient Descent Updates

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \alpha \text{del} J(\boldsymbol{\beta})$$

Batch: full n; SGD: 1 sample;

Mini-batch: size B (e.g. 32, 64).

Logistic Classification

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\text{Odds} = \frac{p}{1-p} = e^z$$

$$\text{Logit}(p) = \ln(\text{Odds}) = \mathbf{w}^T\mathbf{x} + b$$

Loss: Binary Cross-Entropy

Important Examination Questions (Regression)

Curated questions structured according to VTU and Ramaiah examination standards.

2-Mark / 3-Mark Questions

Practice Problem:

Q1. Differentiate between Linear Regression and Logistic Regression.

Solution:

Linear Regression: Models continuous numerical targets $y \in \mathbb{R}$ using unbounded linear hypotheses $\hat{y} = \mathbf{w}^T\mathbf{x} + b$, evaluated via MSE.

Logistic Regression: Models categorical class probabilities $P(y = 1|\mathbf{x}) \in (0, 1)$ using the non-linear Sigmoid activation function, evaluated via Binary Cross-Entropy (Log-Loss).

Practice Problem:

Q2. Define the Sigmoid function and state its range and mathematical derivative.

Solution:

The Sigmoid activation function is defined as $\sigma(z) = \frac{1}{1+e^{-z}}$. It squashes any real-valued input from $(-\infty, +\infty)$ into the probability range $(0, 1)$. Its derivative is computationally efficient: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.

Practice Problem:

Q3. Why is Adjusted R^2 preferred over standard R^2 in Multiple Regression?

Solution:

Standard R^2 mechanically increases or remains constant whenever a new predictor is added to a model, even if the feature is useless random noise. Adjusted R^2 introduces a degrees-of-freedom penalty for every additional parameter added: $R^2_{\text{adj}} = 1 - \left[\frac{(1-R^2)(n-1)}{n-p-1}\right]$, ensuring it increases only if the new feature genuinely improves predictive accuracy.

5-Mark / 6-Mark Questions

Practice Problem:

Q4. Compare Ridge Regression (L2) and Lasso Regression (L1) with their loss functions and properties.

Solution:

Ridge Regression: Adds an L_2 squared penalty $\lambda \sum \beta_j^2$. Has closed-form solution $\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$. Shrinks coefficients smoothly toward zero but never forces them to exact zero.

Lasso Regression: Adds an L_1 absolute value penalty $\lambda \sum |\beta_j|$. Due to the sharp corners of the L_1 constraint diamond, it forces irrelevant coefficients to become identically zero, performing automatic embedded feature selection.

Practice Problem:

Q5. Explain the concept of Odds Ratio and Log-Odds (Logit function) in Logistic Regression.

Solution:

Odds: The ratio of probability of occurrence to probability of non-occurrence: $\text{Odds} = \frac{p}{1-p} = e^z$.

Logit: Taking the natural logarithm of odds yields the logit function: $\text{Logit}(p) = \ln\left(\frac{p}{1-p}\right) = z = \mathbf{w}^T\mathbf{x} + b$.

This shows that Logistic Regression is a generalized linear model that assumes the log-odds of the positive class are a linear combination of the input predictors.

8-Mark / 10-Mark Questions

Practice Problem:

Q6. Derive the Ordinary Least Squares (OLS) closed-form formulas for slope b1 and intercept b0 in simple linear regression.

Solution:

Objective: Minimize the Sum of Squared Residuals: $S(\beta_0, \beta_1) = \sum_{i=1}^{n}(y_i - (\beta_0 + \beta_1 x_i))^2$.

1. Partial derivative with respect to β_0:

$\frac{\partial S}{\partial \beta_0} = -2\sum(y_i - \beta_0 - \beta_1 x_i) = 0 \implies \sum y_i - n\beta_0 - \beta_1 \sum x_i = 0$

Dividing by n: $\bar{y} - \beta_0 - \beta_1 \bar{x} = 0 \implies \boldsymbol{\beta_0 = \bar{y} - \beta_1 \bar{x}}$.

2. Partial derivative with respect to β_1:

$\frac{\partial S}{\partial \beta_1} = -2\sum x_i(y_i - \beta_0 - \beta_1 x_i) = 0 \implies \sum x_i y_i - \beta_0 \sum x_i - \beta_1 \sum x_i^2 = 0$

Substitute $\beta_0 = \bar{y} - \beta_1 \bar{x}$:

$\sum x_i y_i - (\bar{y} - \beta_1 \bar{x})\sum x_i - \beta_1 \sum x_i^2 = 0 \implies \sum x_i y_i - \bar{y}\sum x_i = \beta_1 \left(\sum x_i^2 - \bar{x}\sum x_i\right)$

Using algebraic identities $\sum x_i y_i - n\bar{x}\bar{y} = \sum(x_i - \bar{x})(y_i - \bar{y})$ and $\sum x_i^2 - n\bar{x}^2 = \sum(x_i - \bar{x})^2$:

$$\beta_1 = \frac{\sum_{i=1}^{n}(\mathbf{x_i} - \bar{\mathbf{x}})(\mathbf{y_i} - \bar{\mathbf{y}})}{\sum_{i=1}^{n}(\mathbf{x_i} - \bar{\mathbf{x}})^2} = \frac{\mathbf{n}\sum \mathbf{x_i y_i} - (\sum \mathbf{x_i})(\sum \mathbf{y_i})}{\mathbf{n}\sum \mathbf{x_i^2} - (\sum \mathbf{x_i})^2}$$

Solved Questions from Past Department Examinations

ARCHIVE

Transcribed directly from official Ramaiah Institute of Technology papers in `notes/ml/practice/ml-cie-1-2-see.pdf` (CIE-1 March 2026, Course Code 23IS62) and faculty coursework:

Ramaiah Internal Assessment – I (March 2026) · Q 1(c) [3 Marks · Blooms L3, CO3]
"The following data represents hours studied (X) and marks obtained (Y): X = [2, 4, 6, 8, 10], Y = [50, 60, 65, 80, 85]. i. Compute slope (b) and intercept (a). ii. Write the regression equation. iii. Predict marks for X = 7 hours."

Step 1: Compute Means ($n = 5$):

$$\bar{X} = \frac{2+4+6+8+10}{5} = \frac{30}{5} = \mathbf{6.0}, \quad \bar{Y} = \frac{50+60+65+80+85}{5} = \frac{340}{5} = \mathbf{68.0}$$

Step 2: Compute Summations:

$$\sum X = 30, \quad \sum Y = 340, \quad \sum X^2 = 220, \quad \sum XY = (2)(50) + (4)(60) + (6)(65)$$

Step 3: Compute Slope (b) and Intercept (a):

$$b = \frac{n\sum XY - (\sum X)(\sum Y)}{n\sum X^2 - (\sum X)^2} = \frac{5(2220) - (30)(340)}{5(220) - (30)^2} = \frac{11100 - 10200}{1100 - 900} = \frac{900}{200} = \mathbf{4.5}$$

$$a = \bar{Y} - b\bar{X} = 68.0 - (4.5000)(6.0) = 68.0 - 27.0 = \mathbf{41.0000}$$

Step 4: Regression Equation & Prediction:

$$\text{Regression Equation: } \hat{Y} = 41.0000 + 4.5000X$$

$$\text{For } X = 7 \text{ hours: } \hat{Y}(7) = 41.0000 + 4.5000(7) = 41.0000 + 31.50 = \mathbf{72.50} \text{ marks}$$

Ramaiah Internal Assessment – I (March 2026) · Q 2(c) [3 Marks · Blooms L2, CO3]
"Differentiate Logistic Regression and Linear Regression in machine learning."

1. **Output Target:** Linear Regression predicts continuous numeric quantities $y \in (-\infty, +\infty)$, whereas Logistic Regression predicts discrete probabilities $P(y = 1|\mathbf{x}) \in (0, 1)$ mapped to class labels $\{0, 1\}$.
2. **Hypothesis Function:** Linear regression uses $\hat{y} = \mathbf{w}^T\mathbf{x} + b$; logistic regression wraps the linear score inside the Sigmoid function $\sigma(z) = 1/(1 + e^{-z})$.
3. **Loss Function:** Linear regression minimizes Mean Squared Error (MSE), whereas logistic regression minimizes Binary Cross-Entropy (Log-Loss) because MSE is non-convex when combined with the sigmoid activation function.

Practice Problems (Step-by-Step Verified Solutions)

PRACTICE

The following problems are drawn from the faculty problem sheets (`Linear_Regression and multiple linear regression_problems.docx` and `Linear & ridge regression....docx`) with verified step-by-step arithmetic:

Problem 1: Simple Linear Regression · Ice Cream Sales vs Temperature Easy · Verified Formula

A shop records daily temperature (°C) and corresponding ice cream sales (units) for 8 days: $X = [20, 22, 24, 26, 28, 30, 32, 34]$, $Y = [120, 135, 150, 165, 180, 195, 210, 225]$. 1. Find the regression equation. 2. Predict sales when temperature is 29°C.

Summations ($n = 8$): $\sum X = 216 \implies \bar{X} = 27.00$, $\sum Y = 1380 \implies \bar{Y} = 172.50$, $\sum X^2 = 6000$, $\sum XY = 38520$.

Slope: $b = \frac{8(38520) - (216)(1380)}{8(6000) - (216)^2} = \frac{308160 - 298080}{48000 - 46656} = \frac{10080}{1344} = \mathbf{7.5000}$.

Intercept: $a = 172.50 - (7.5000)(27.00) = 172.50 - 202.50 = \mathbf{-30.0000}$.

Equation: $\hat{Y} = -30.0000 + 7.5000X$. For $X = 29°\text{C}$: $\hat{Y}(29) = -30.0 + 7.5(29) = \mathbf{187.50}$ units.

Problem 2: Regularization · Ridge vs OLS Coefficient Shrinkage Moderate · Verified Matrix Math

Given centered 1D feature data where $\sum x_i^2 = 100$ and $\sum x_i y_i = 400$. Calculate the OLS slope and the Ridge slope for regularization penalty $\lambda = 20$.

OLS Slope: $\beta_{\text{OLS}} = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{400}{100} = \mathbf{4.0000}$.

Ridge Slope ($\lambda = 20$): $\beta_{\text{Ridge}} = \frac{\sum x_i y_i}{\sum x_i^2 + \lambda} = \frac{400}{100 + 20} = \frac{400}{120} = \mathbf{3.3333}$.

Conclusion: The Ridge penalty successfully shrunk the coefficient from 4.00 to 3.33 (a 16.7% shrinkage), dampening sensitivity to potential noise in the feature.